

Advanced Timer IP Design Specification

RTL Architecture, Logic Schematics, Timing Waveforms,
and Verification Report

Top Module: `timer_top`

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Project Level: **Advanced Level**

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1 Overview

1.1 Introduction

Timers are essential peripheral modules in modern digital and embedded architectures, providing precise time-base intervals, event counting, pulse generation, delay measurement, and interrupt servicing.

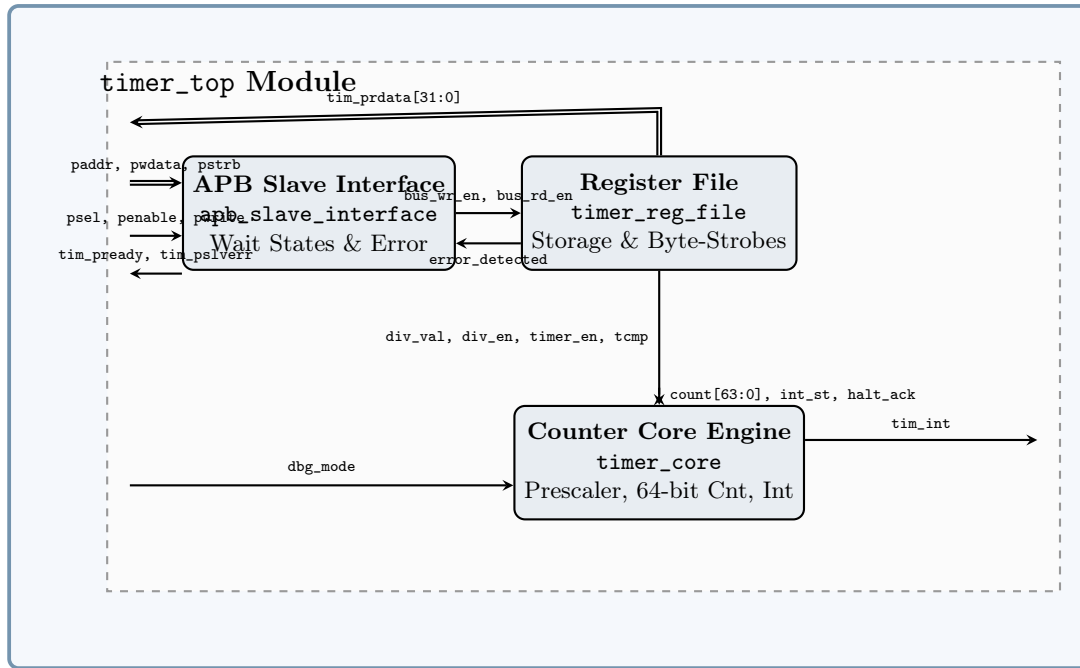
In this project, a custom 64-bit Advanced Timer IP (`timer_top`) is developed based on the **Core Local Interruptor (CLINT)** architecture of RISC-V processors. The peripheral integrates seamlessly as a slave module on the **AMBA Advanced Peripheral Bus (APB)**.

1.2 Main Features

- **64-bit Count-Up Core:** High-capacity continuous counter accessible via 32-bit registers (`TDR0` and `TDR1`).
- **12-bit Address Map:** 4KB memory space (`0x000–0xFFF`) with Read-As-Zero / Write-Ignored (RAZ/WI) support for unmapped offsets.
- **APB Slave with 1-Cycle Wait State:** Compliant slave asserting `tim_pready` after a 1-cycle access delay to relax critical timing paths.
- **Byte-Strobe Write Access:** Fine-grained byte-level write capability across all registers using `tim_pstrb[3:0]`.
- **Hardware Error Handling (`tim_pslverr`):** Automatic bus error generation when prohibited operations are attempted, blocking unauthorized register corruption.
- **Multi-Mode Prescaler:** Divides system clock frequency by 2^0 to 2^8 (scale factors: 1, 2, 4, 8, 16, 32, 64, 128, 256).
- **Debug Mode Halt Handshake:** Synchronously freezes the counter and prescaler upon request (`halt_req`) while in debug mode (`dbg_mode`), preserving sub-cycle phase timing.
- **Hardware Auto-Clear on Disable:** Hardware resets `TDR0` and `TDR1` to `32'h0` on a $1 \rightarrow 0$ falling edge transition of `timer_en`.
- **Maskable Interrupt:** Active-high level-triggered interrupt output (`tim_int`) with Write-1-to-Clear (RW1C) status in `TISR`.

1.3 Block Diagram

The top-level Timer IP is partitioned into three decoupled functional blocks interconnected inside `timer_top`, as shown in Figure 1.

Figure 1: System Block Diagram of the Timer IP (`timer_top`)

1.4 Interface Signals

Table 1 lists the complete pinout of the `timer_top` module.

Table 1: Interface Signal Definitions of `timer_top`

Signal Name	Width	Direction	Active	Description
<code>sys_clk</code>	1	Input	Rising	System clock for all internal logic.
<code>sys_rst_n</code>	1	Input	Low	Asynchronous active-low global reset.
<code>tim_psel</code>	1	Input	High	APB select signal indicating the timer is addressed.
<code>tim_pwrite</code>	1	Input	High	APB transfer direction: 1 = Write, 0 = Read.
<code>tim_penable</code>	1	Input	High	APB enable signal indicating the access phase.
<code>tim_paddr[11:0]</code>	12	Input	—	APB address bus (0x000 to 0xFFF).
<code>tim_pwdata[31:0]</code>	32	Input	—	APB write data bus driven by the host.
<code>tim_pstrb[3:0]</code>	4	Input	High	Byte lane write strobes for byte-level access.
<code>tim_prdata[31:0]</code>	32	Output	—	APB read data bus returning register data.
<code>tim_pready</code>	1	Output	High	APB ready response (1-cycle wait state).
<code>tim_pslverr</code>	1	Output	High	APB error response on prohibited access.
<code>tim_int</code>	1	Output	High	Active-high level-triggered interrupt output.
<code>dbg_mode</code>	1	Input	High	Debug mode indicator from host system.

2 Register Specification

2.1 Register Summary

Table 2 summarizes the memory map of the Timer IP.

Table 2: Register Summary

Address	Item	Type	Reset Value	Register Name & Function
0x00	TCR	RW	32'h0000_0100	Timer Control Register
0x04	TDR0	RW	32'h0000_0000	Timer Data Register 0 (Lower 32-bit Counter)
0x08	TDR1	RW	32'h0000_0000	Timer Data Register 1 (Upper 32-bit Counter)
0x0C	TCMP0	RW	32'hFFFF_FFFF	Timer Compare Register 0 (Lower 32-bit Compare)
0x10	TCMP1	RW	32'hFFFF_FFFF	Timer Compare Register 1 (Upper 32-bit Compare)
0x14	TIER	RW	32'h0000_0000	Timer Interrupt Enable Register
0x18	TISR	RW1C	32'h0000_0000	Timer Interrupt Status Register
0x1C	THCSR	Mixed	32'h0000_0000	Timer Halt Control Status Register
<i>Others</i>	<i>Reserved</i>	RAZ/WI	32'h0000_0000	Unmapped space (Read 0, Write Ignored)

2.2 Timer Control Register (TCR) — 0x00

Reset Value: 32'h0000_0100

Table 3: Timer Control Register (TCR) Bitfields

Bit	Name	Type	Default	Description
31:12	Reserved	RO	20'h0	Reserved. Read as 0.
11:8	div_val	RW	4'b0001	Prescaler Divisor Setting: • 4'b0000: Divide by 1 (No division) • 4'b0001: Divide by 2 (Default reset value) • 4'b0010: Divide by 4 • 4'b0011: Divide by 8 • 4'b0100: Divide by 16 • 4'b0101: Divide by 32 • 4'b0110: Divide by 64 • 4'b0111: Divide by 128 • 4'b1000: Divide by 256 • <i>Others (4'b1001–4'b1111):</i> Prohibited setting. Generates <code>tim_pslverr</code> and write is ignored. Constraint: Modifying <code>div_val</code> while <code>timer_en = 1</code> is prohibited and generates <code>tim_pslverr</code> .
7:2	Reserved	RO	6'h0	Reserved. Read as 0.
1	div_en	RW	1'b0	Counter Control Mode Enable: • 0: Disabled. Counter runs at system clock speed. • 1: Enabled. Counter speed is determined by <code>div_val</code>. Constraint: Modifying <code>div_en</code> while <code>timer_en = 1</code> is prohibited and generates <code>tim_pslverr</code> .
0	timer_en	RW	1'b0	Timer Enable: • 0: Disabled. Counter does not increment. • 1: Enabled. Counter starts incrementing. Hardware Auto-Clear: Transitioning <code>1 → 0</code> resets TDR0 and TDR1 to 32'h0.

2.3 Timer Data Register 0 (TDR0) — 0x04

Reset Value: 32'h0000_0000

Table 4: Timer Data Register 0 (TDR0) Bitfields

Bit	Name	Type	Default	Description
31:0	TDR0	RW	32'h0000_0000	Lower 32-bit of the 64-bit counter (<code>count[31:0]</code>). Automatically cleared to 32'h0 when <code>timer_en</code> changes from High to Low.

2.4 Timer Data Register 1 (TDR1) — 0x08

Reset Value: 32'h0000_0000

Table 5: Timer Data Register 1 (TDR1) Bitfields

Bit	Name	Type	Default	Description
31:0	TDR1	RW	32'h0000_0000	Upper 32-bit of the 64-bit counter (<code>count[63:32]</code>). Automatically cleared to 32'h0 when <code>timer_en</code> changes from High to Low.

2.5 Timer Compare Register 0 (TCMP0) — 0x0C

Reset Value: 32'hFFFFFF_FFFF

Table 6: Timer Compare Register 0 (TCMP0) Bitfields

Bit	Name	Type	Default	Description
31:0	TCMP0	RW	32'hFFFFFF_FFFF	Lower 32-bit of the 64-bit compare value. When <code>count == {TCMP1, TCMP0}</code> , the interrupt status bit is asserted.

2.6 Timer Compare Register 1 (TCMP1) — 0x10

Reset Value: 32'hFFFFFF_FFFF

Table 7: Timer Compare Register 1 (TCMP1) Bitfields

Bit	Name	Type	Default	Description
31:0	TCMP1	RW	32'hFFFFFF_FFFF	Upper 32-bit of the 64-bit compare value. When <code>count == {TCMP1, TCMP0}</code> , the interrupt status bit is asserted.

2.7 Timer Interrupt Enable Register (TIER) — 0x14

Reset Value: 32'h0000_0000

Table 8: Timer Interrupt Enable Register (TIER) Bitfields

Bit	Name	Type	Default	Description
31:1	Reserved	RO	31'h0	Reserved. Read as 0.
0	int_en	RW	1'b0	Timer Interrupt Enable: • 0: Disabled. <code>tim_int</code> output is masked to 0. • 1: Enabled. <code>tim_int</code> is driven High when <code>TISR.int_st = 1</code>. Note: Clearing this bit to 0 masks <code>tim_int</code> but does not alter <code>TISR.int_st</code> .

2.8 Timer Interrupt Status Register (TISR) — 0x18

Reset Value: 32'h0000_0000

Table 9: Timer Interrupt Status Register (TISR) Bitfields

Bit	Name	Type	Default	Description
31:1	Reserved	RO	31'h0	Reserved. Read as 0.
0	int_st	RW1C	1'b0	Interrupt Pending Status Bit: • 0: Match trigger condition has not occurred. • 1: Match condition occurred (<code>count == tcmp</code>). Clearance Rule: Write 1 to clear to 0. Writing 0 has no effect. Write clearance has highest priority over simultaneous set triggers.

2.9 Timer Halt Control Status Register (THCSR) — 0x1C

Reset Value: 32'h0000_0000

Table 10: Timer Halt Control Status Register (THCSR) Bitfields

Bit	Name	Type	Default	Description
31:2	Reserved	RO	30'h0	Reserved. Read as 0.
1	halt_ack	RO	1'b0	Halt Acknowledge Status: • 0: Timer is running or debug mode is inactive. • 1: Timer is halted in debug mode. Evaluates combinationally as: <code>halt_ack = halt_req & dbg_mode</code> .
0	halt_req	RW	1'b0	Halt Request: • 0: Normal counter operation. • 1: Requests timer to halt in debug mode.

3 Functional Description

3.1 APB Slave and Register File Architecture

The APB Slave block implements standard APB protocol handshaking with support for 1-cycle wait-states, byte-strobe write masking, read multiplexing, and prohibited-access error decoding.

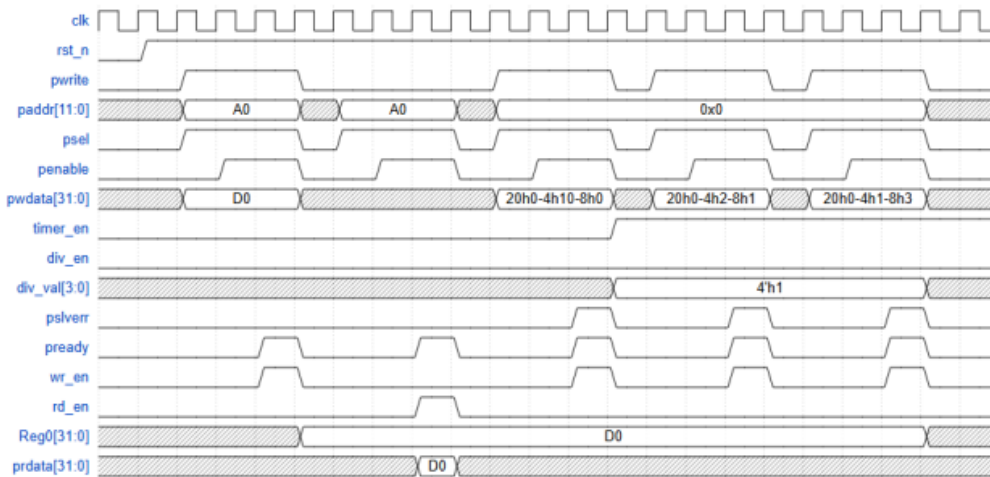


Figure 2: APB Slave Transfer Timing with 1-Cycle Wait State

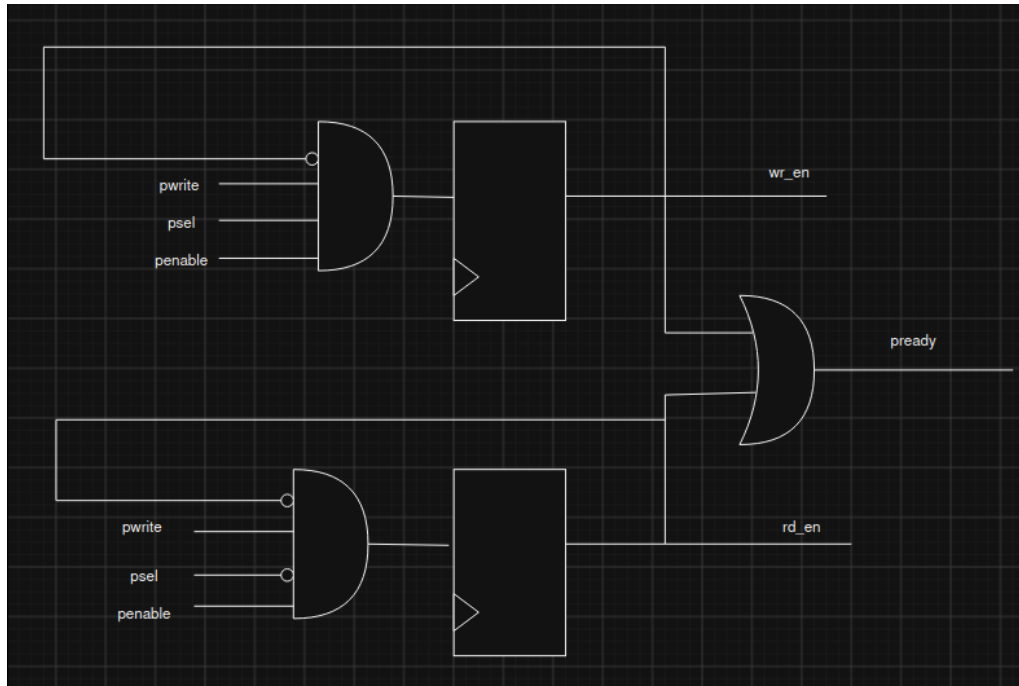


Figure 3: APB Slave Interface Logic Diagram

3.1.1 Timer Control Register (TCR) Logic

Figure 4 illustrates the write decoding and error generation logic for the TCR register. When an APB write targets address `0x00`, write data is gated by `pstrb[0]` and `pstrb[1]`. If a prohibited divisor value is supplied or if software attempts to change divisor settings while `timer_en = 1`, `pslverr` is asserted and the register update is blocked.

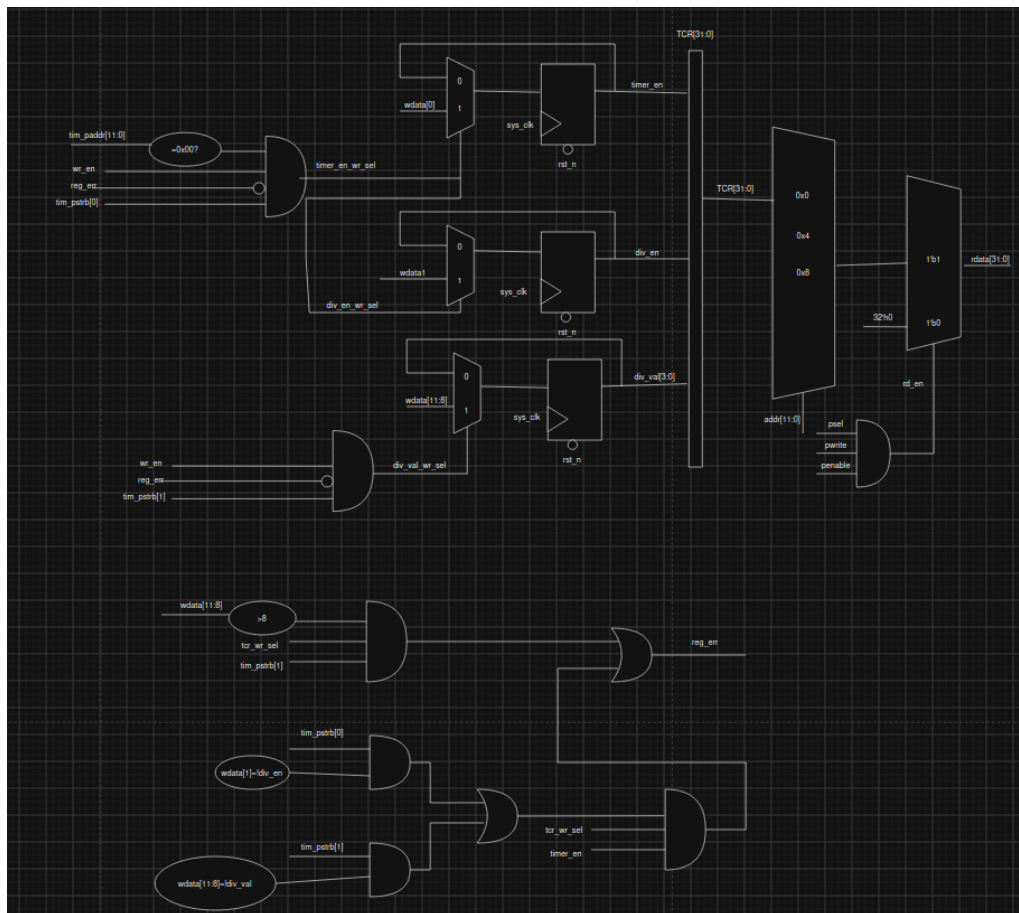


Figure 4: TCR Register Logic Diagram

3.1.2 Timer Data Registers (TDR0 / TDR1) Logic

Figures 5 and 6 show the next-state logic for TDR0 and TDR1.

- Direct APB writes via `pstrb[3:0]` have highest priority.
- A falling edge on `timer_en` (`timer_en_neg`) resets both registers to `32'h0000_0000`.
- When counting is enabled (`count_en = 1`), TDR0 and TDR1 capture the 64-bit incremented value `count + 1`.

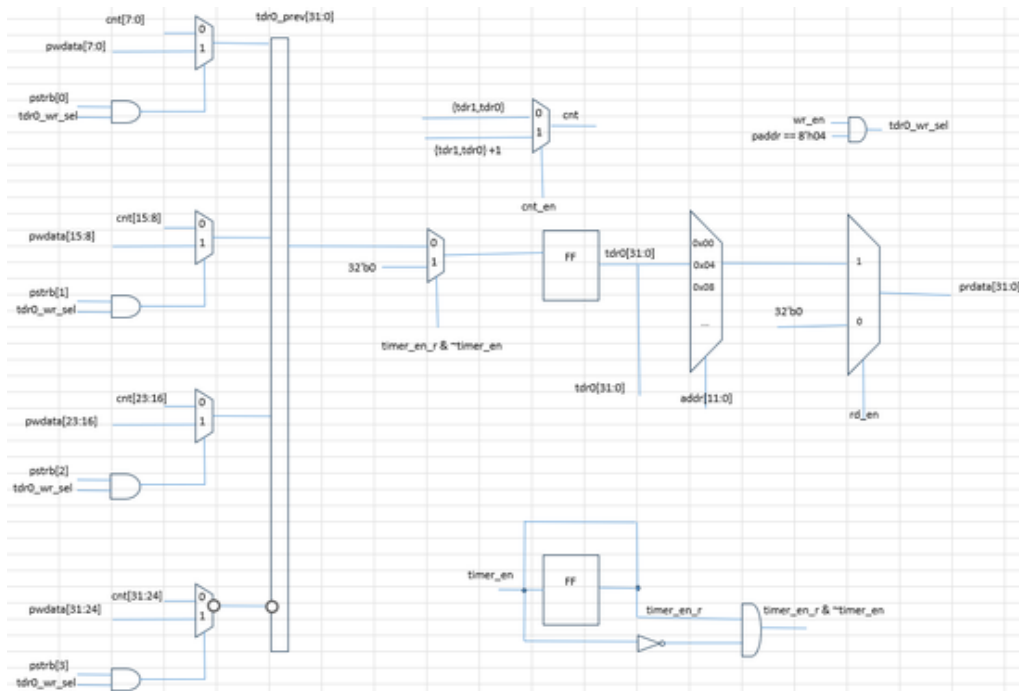


Figure 5: TDR0 Register Logic Diagram

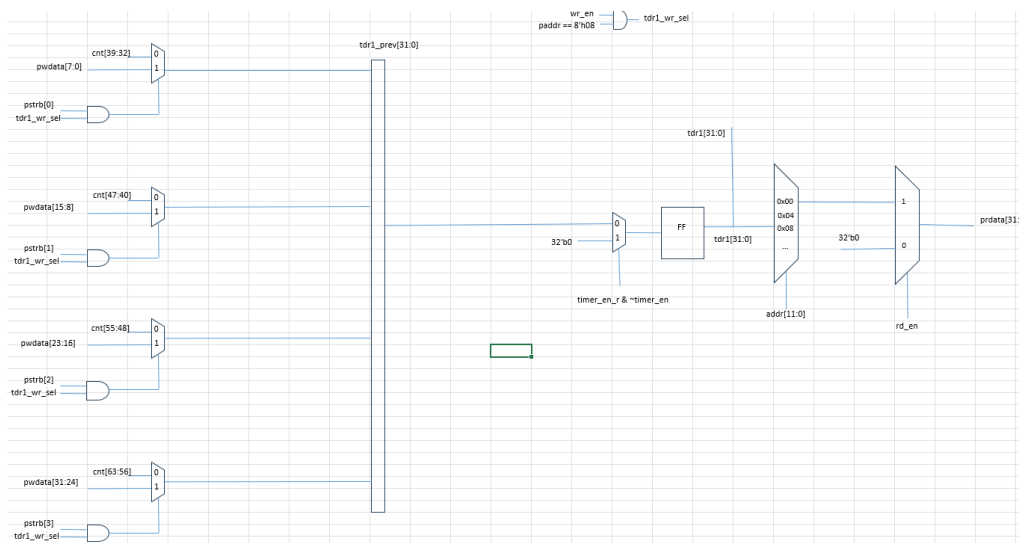


Figure 6: TDR1 Register Logic Diagram

3.1.3 Timer Compare Registers (TCMP0 / TCMP1) Logic

Figures 7 and 8 show the byte-strobe storage logic for the 64-bit compare registers (TCMP0 at 0x0C and TCMP1 at 0x10).

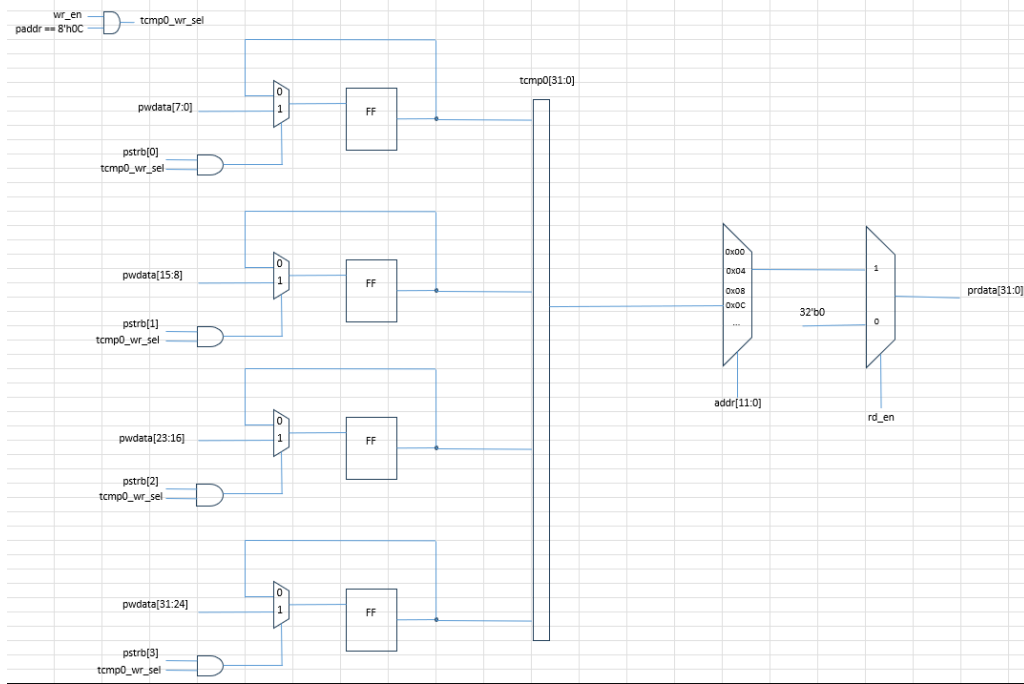


Figure 7: TCMP0 Register Logic Diagram

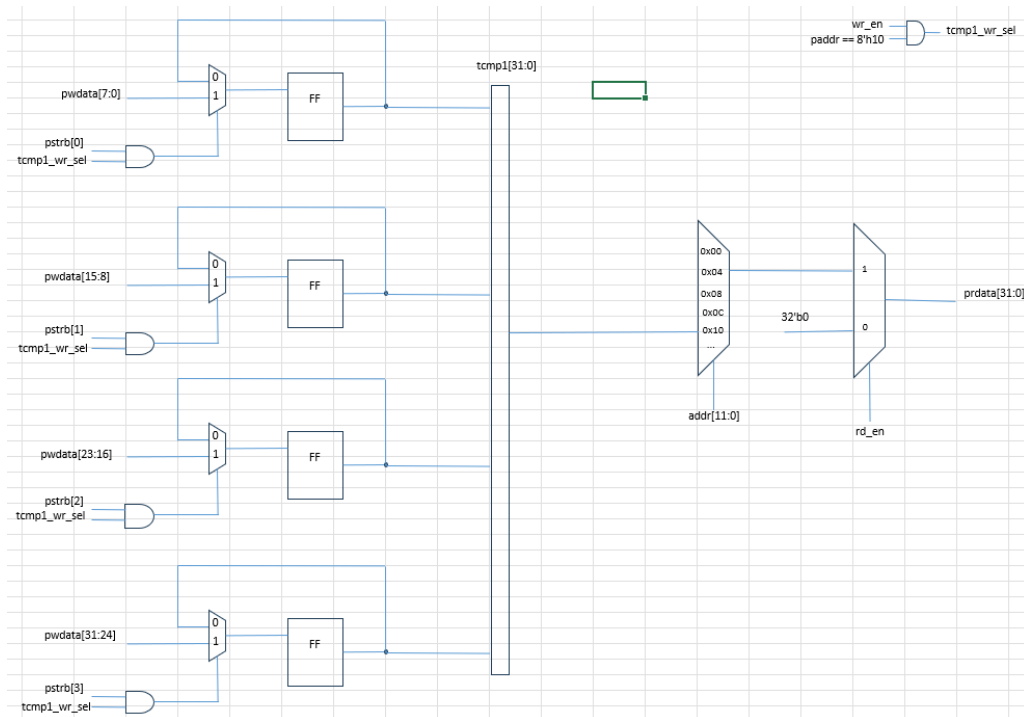


Figure 8: TCMP1 Register Logic Diagram

3.1.4 Interrupt Enable and Status (TIER / TISR) Logic

Figure 9 shows the interrupt generation logic:

- When `count == tcmp`, `int_set` is asserted to latch `TISR.int_st = 1`.
- Writing 1 to bit 0 of TISR asserts `int_clr`, which clears `int_st` to 0. Clearance takes highest priority.

- `tim_int` is driven as `tim_int = int_st & int_en`.

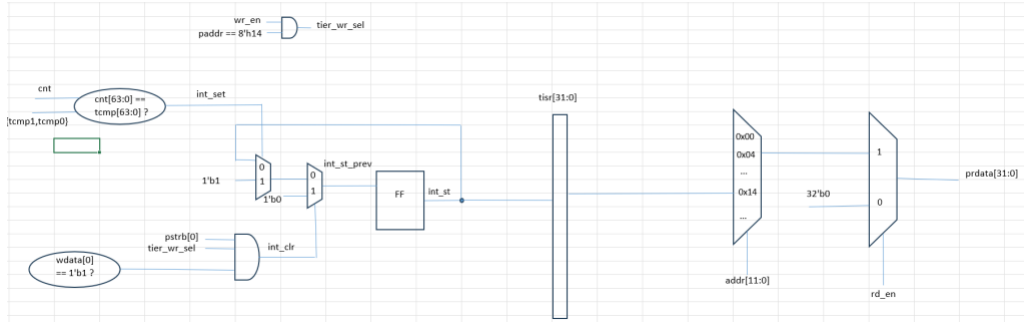


Figure 9: TIER and TISR Register Logic Diagram

3.1.5 Timer Halt Control Status Register (THCSR) Logic

Figure 10 illustrates the debug halt logic: `halt_req` is writable via APB, and `halt_ack` acknowledges the halt state combinationally when `dbg_mode = 1`.

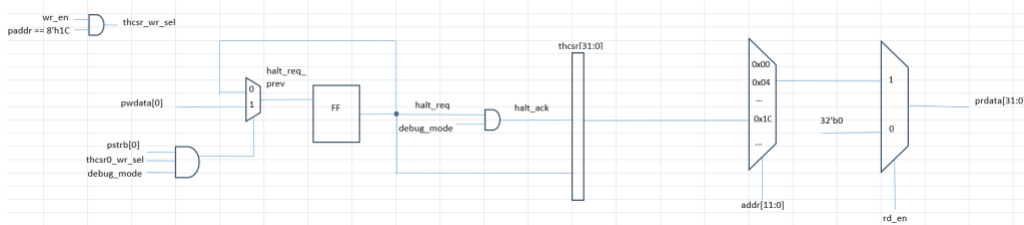


Figure 10: THCSR Register Logic Diagram

3.2 Counter Control, Prescaler Modes, and Waveforms

Figure 11 displays the Counter Control schematic (`cnt_ctrl1`) that decodes `div_val` into prescaler limits and pulses `count_en`.

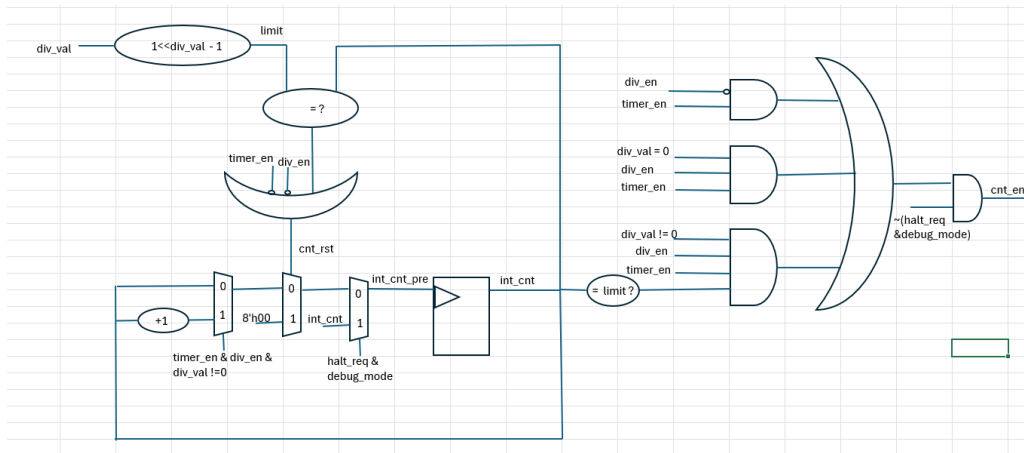


Figure 11: Counter Control Logic Diagram

The following subsections describe the 10 operational timing modes and waveforms.

3.2.1 Default Mode Counting Waveform ($\text{div_en} = 0$)

In default mode, the counter increments on every rising edge of `sys_clk` when `timer_en` = 1. When `dbg_mode` = 1 and `halt_req` = 1, the counter freezes and resumes counting once unhalted.

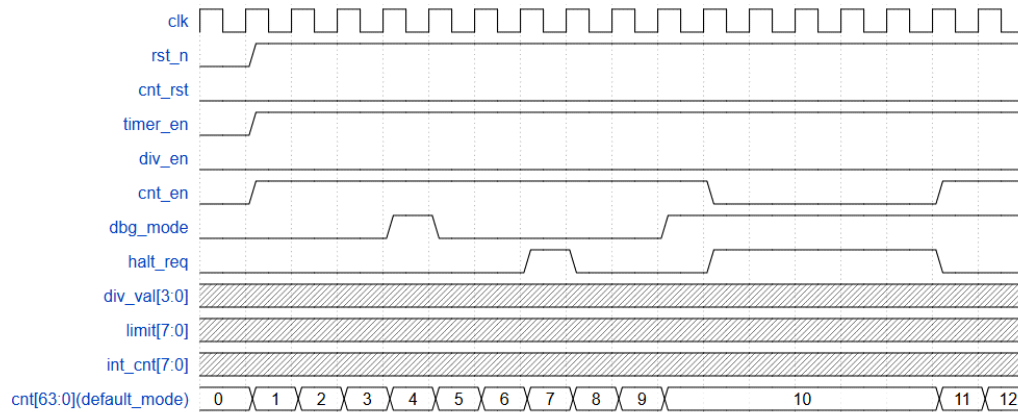


Figure 12: Waveform of Counter in Default Mode ($\text{div_en} = 0$)

3.2.2 Control Mode — $\text{div_val} = 0$ (Divide by 1)

When $\text{div_en} = 1$ and $\text{div_val} = 0$, the counter operates at undivided clock speed.

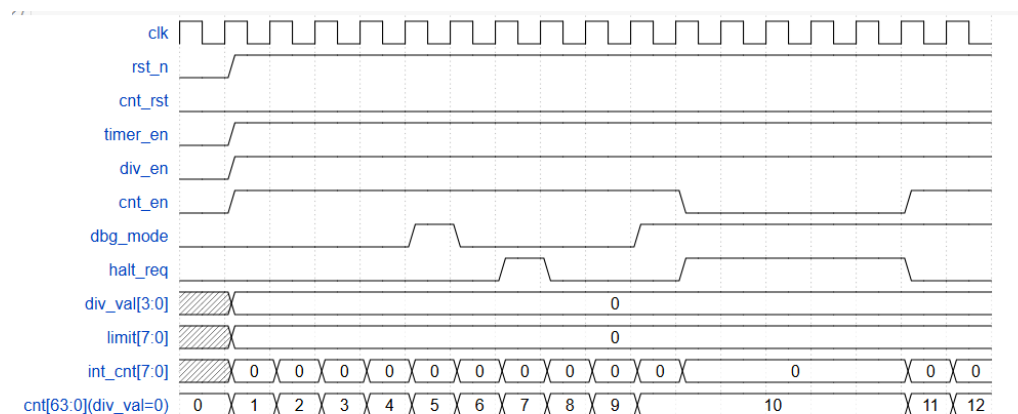


Figure 13: Waveform of Counter in Control Mode ($\text{div_val} = 0$)

3.2.3 Control Mode — $\text{div_val} = 1$ (Divide by 2)

The counter increments once every 2 system clock cycles ($\text{limit} = 1$). During debug halt, the prescaler `int_cnt` freezes its state, ensuring timing phase preservation upon resume.

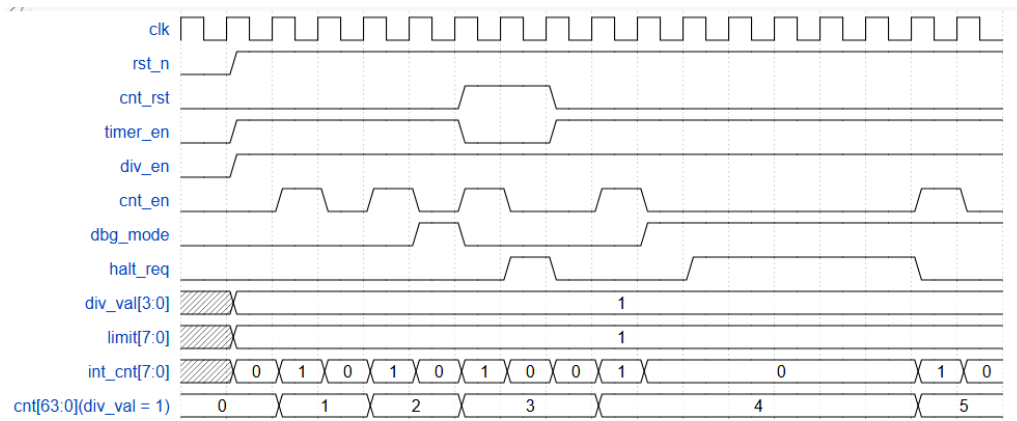


Figure 14: Waveform of Counter in Divide-by-2 Mode (`div_val = 1`)

3.2.4 Control Mode — div_val = 2 (Divide by 4)

The counter increments once every 4 clock cycles (limit = 3).

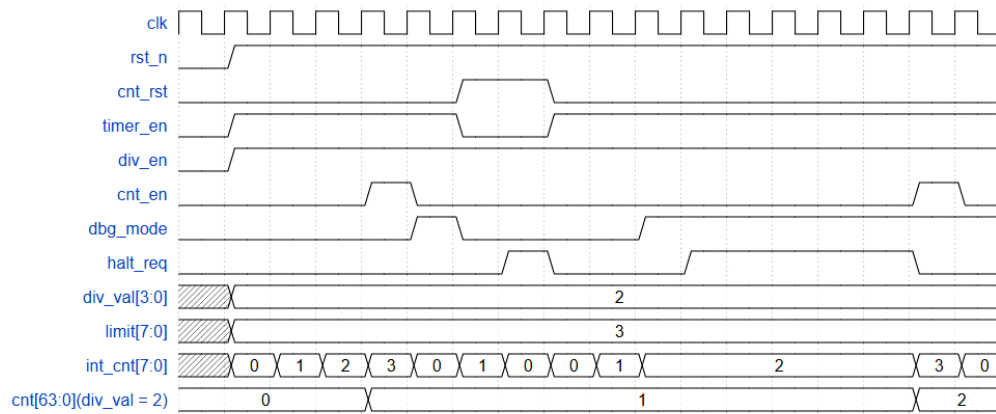


Figure 15: Waveform of Counter in Divide-by-4 Mode (`div_val = 2`)

3.2.5 Control Mode — div_val = 3 (Divide by 8)

The counter increments once every 8 clock cycles (limit = 7).

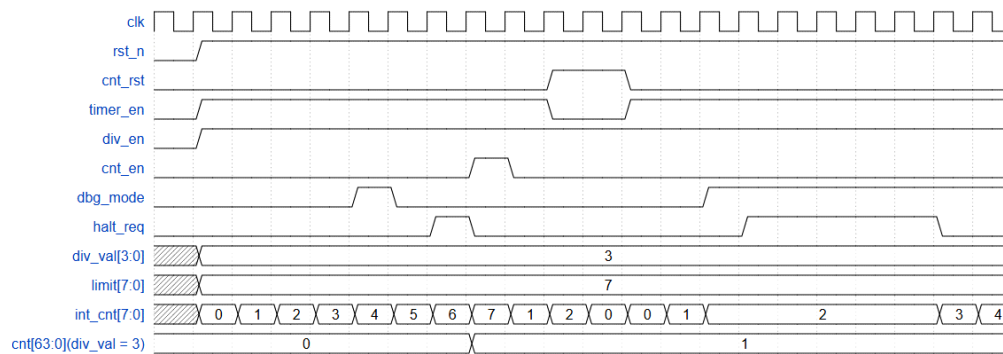


Figure 16: Waveform of Counter in Divide-by-8 Mode (`div_val` = 3)

3.2.6 Control Mode — $\text{div_val} = 4$ (Divide by 16)

The counter increments once every 16 clock cycles (limit = 15).

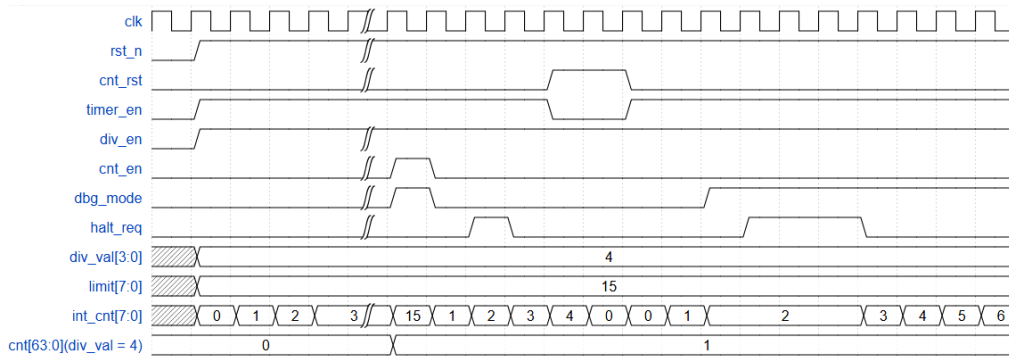


Figure 17: Waveform of Counter in Divide-by-16 Mode ($\text{div_val} = 4$)

3.2.7 Control Mode — $\text{div_val} = 5$ (Divide by 32)

The counter increments once every 32 clock cycles (limit = 31).

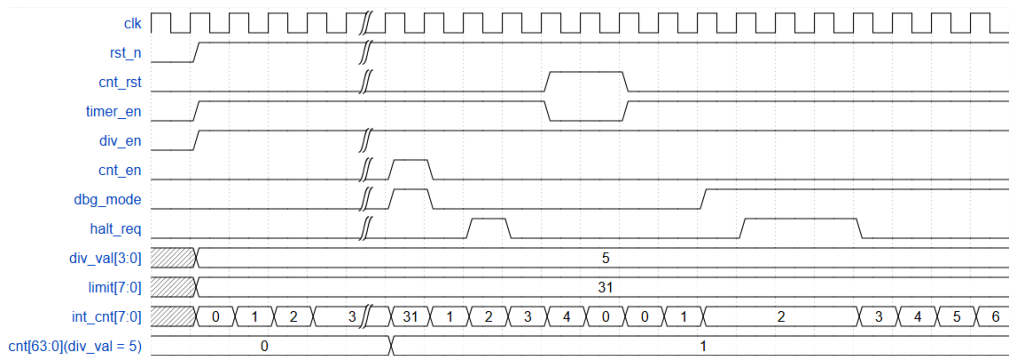


Figure 18: Waveform of Counter in Divide-by-32 Mode ($\text{div_val} = 5$)

3.2.8 Control Mode — $\text{div_val} = 6$ (Divide by 64)

The counter increments once every 64 clock cycles (limit = 63).

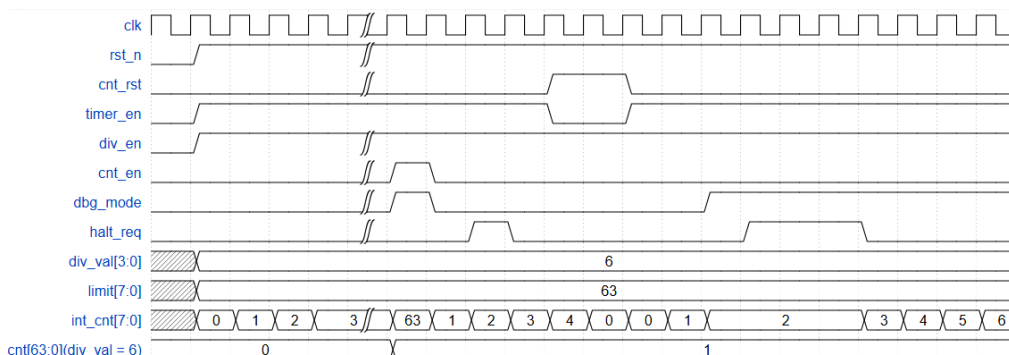


Figure 19: Waveform of Counter in Divide-by-64 Mode ($\text{div_val} = 6$)

3.2.9 Control Mode — $\text{div_val} = 7$ (Divide by 128)

The counter increments once every 128 clock cycles (limit = 127).

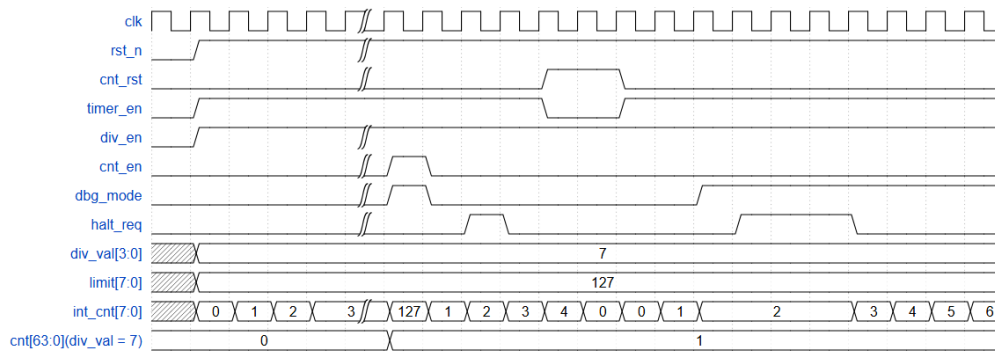


Figure 20: Waveform of Counter in Divide-by-128 Mode ($\text{div_val} = 7$)

3.2.10 Control Mode — $\text{div_val} = 8$ (Divide by 256)

The counter increments once every 256 clock cycles (limit = 255).

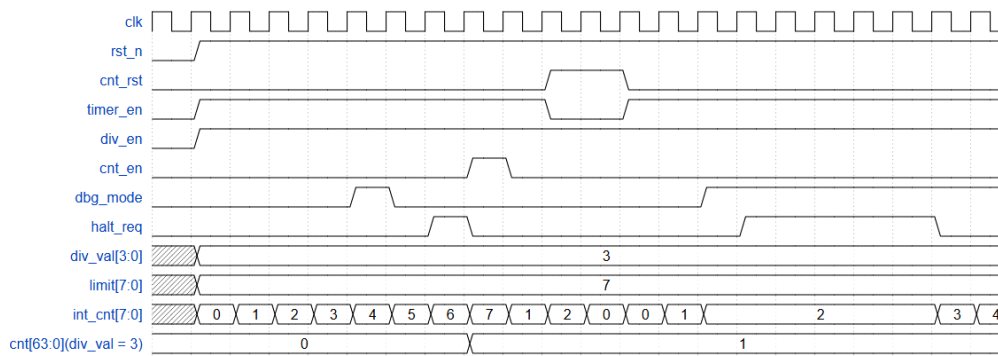


Figure 21: Waveform of Counter in Divide-by-256 Mode ($\text{div_val} = 8$)

4 Synthesizable Verilog RTL Implementation

4.1 apb_slave_interface.v

```

1 module apb_slave_interface (
2     input wire      sys_clk,
3     input wire      sys_rst_n,
4     input wire      tim_psel,
5     input wire      tim_penable,
6     input wire      tim_pwrite,
7     output wire     tim_pready,
8     output wire     tim_pslverr,
9     input wire      error_detected,
10    output wire     bus_wr_en,
11    output wire     bus_rd_en
12 );
13 reg wen, ren;
14 always @(posedge sys_clk or negedge sys_rst_n) begin
15     if (!sys_rst_n) begin
16         wen <= 1'b0;
17         ren <= 1'b0;

```

```

18     end else begin
19         wen <= tim_psel & tim_pwrite & tim_penable & ~wen;
20         ren <= tim_psel & ~tim_pwrite & tim_penable & ~ren;
21     end
22 end
23
24 assign bus_wr_en    = wen & !error_detected;
25 assign bus_rd_en    = ren;
26 assign tim_pready   = wen | ren;
27 assign tim_pslverr  = error_detected && (wen | ren);
28 endmodule

```

Listing 1: APB Slave Interface (apb_slave_interface.v)

4.2 timer_reg_file.v

```

1 module timer_reg_file (
2     input wire      sys_clk,
3     input wire      sys_rst_n,
4     input wire      tim_psel,
5     input wire      tim_penable,
6     input wire      tim_pwrite,
7     input wire      bus_wr_en,
8     input wire      bus_rd_en,
9     input wire [11:0] tim_paddr,
10    input wire [31:0] tim_pwdata,
11    input wire [3:0] tim_pstrb,
12    output reg [31:0] tim_prdata,
13    output wire      error_detected,
14    output reg [3:0] div_val,
15    output reg      div_en,
16    output reg      timer_en,
17    output wire      timer_en_neg,
18    output wire [63:0] tcmp,
19    output reg      int_en,
20    output wire      int_clr_pulse,
21    output reg      halt_req,
22    input wire [63:0] count,
23    input wire      int_st,
24    input wire      halt_ack,
25    output wire      tdr0_wr_en,
26    output wire      tdr1_wr_en,
27    output wire [31:0] tdr_wr_data,
28    output wire [3:0] tdr_wr_strb
29 );
30 wire is_tcr_sel = (tim_paddr == 12'h00);
31 wire div_val_err = is_tcr_sel && (tim_pwdata[11:8] > 4'd8) && tim_pstrb[1];
32 wire div_val_tim_en_err = is_tcr_sel && (tim_pwdata[11:8] != div_val) && tim_pstrb[1] && timer_en;
33 wire div_en_tim_en_err = is_tcr_sel && (tim_pwdata[1] != div_en) && tim_pstrb[0] && timer_en;
34
35 assign error_detected = tim_pwrite && (div_val_err || div_val_tim_en_err ||
36 div_en_tim_en_err);
37
38 reg [31:0] tcmp0, tcmp1;
39 assign tcmp = {tcmp1, tcmp0};
40
41 reg timer_en_1d;
42 always @(posedge sys_clk or negedge sys_rst_n) begin
43     if (!sys_rst_n) timer_en_1d <= 1'b0;
44     else timer_en_1d <= timer_en;
45 end
46 assign timer_en_neg = ~timer_en & timer_en_1d;
47
48 always @(posedge sys_clk or negedge sys_rst_n) begin
49     if (!sys_rst_n) begin
50         div_val <= 4'b0001;
51         div_en <= 1'b0;
52         timer_en <= 1'b0;
53         tcmp0 <= 32'hFFFF_FFFF;

```

```

53     tcmp1    <= 32'hFFFF_FFFF;
54     int_en   <= 1'b0;
55     halt_req <= 1'b0;
56   end else if (bus_wr_en) begin
57     if (tim_paddr == 12'h00) begin
58       if (tim_pstrb[1]) div_val <= tim_pwdata[11:8];
59       if (tim_pstrb[0]) begin
60         div_en <= tim_pwdata[1];
61         timer_en <= tim_pwdata[0];
62       end
63     end
64     if (tim_paddr == 12'h0C) begin
65       if (tim_pstrb[0]) tcmp0[7:0] <= tim_pwdata[7:0];
66       if (tim_pstrb[1]) tcmp0[15:8] <= tim_pwdata[15:8];
67       if (tim_pstrb[2]) tcmp0[23:16] <= tim_pwdata[23:16];
68       if (tim_pstrb[3]) tcmp0[31:24] <= tim_pwdata[31:24];
69     end
70     if (tim_paddr == 12'h10) begin
71       if (tim_pstrb[0]) tcmp1[7:0] <= tim_pwdata[7:0];
72       if (tim_pstrb[1]) tcmp1[15:8] <= tim_pwdata[15:8];
73       if (tim_pstrb[2]) tcmp1[23:16] <= tim_pwdata[23:16];
74       if (tim_pstrb[3]) tcmp1[31:24] <= tim_pwdata[31:24];
75     end
76     if (tim_paddr == 12'h14 && tim_pstrb[0]) int_en <= tim_pwdata[0];
77     if (tim_paddr == 12'h1C && tim_pstrb[0]) halt_req <= tim_pwdata[0];
78   end
79 end
80
81 assign int_clr_pulse = bus_wr_en && (tim_paddr == 12'h18) && tim_pstrb[0] &&
tim_pwdata[0];
82 assign tdr0_wr_en    = bus_wr_en && (tim_paddr == 12'h04);
83 assign tdr1_wr_en    = bus_wr_en && (tim_paddr == 12'h08);
84 assign tdr_wr_data   = tim_pwdata;
85 assign tdr_wr_strb   = tim_pstrb;
86
87 always @(*) begin
88   if (tim_psel && tim_penable && !tim_pwrite) begin
89     case (tim_paddr)
90       12'h00: tim_prdata = {20'h0, div_val, 6'h0, div_en, timer_en};
91       12'h04: tim_prdata = count[31:0];
92       12'h08: tim_prdata = count[63:32];
93       12'h0C: tim_prdata = tcmp0;
94       12'h10: tim_prdata = tcmp1;
95       12'h14: tim_prdata = {31'h0, int_en};
96       12'h18: tim_prdata = {31'h0, int_st};
97       12'h1C: tim_prdata = {30'h0, halt_ack, halt_req};
98       default: tim_prdata = 32'h0000_0000;
99     endcase
100   end else begin
101     tim_prdata = 32'h0000_0000;
102   end
103 end
104 endmodule

```

Listing 2: Register File Module (timer_reg_file.v)

4.3 timer_core.v

```

1 module timer_core (
2   input wire      sys_clk,
3   input wire      sys_rst_n,
4   input wire [3:0] div_val,
5   input wire      div_en,
6   input wire      timer_en,
7   input wire      timer_en_neg,
8   input wire [63:0] tcmp,
9   input wire      int_en,
10  input wire      int_clr_pulse,
11  input wire      halt_req,
12  input wire      dbg_mode,
13  output wire      tim_int,

```

```

14 output wire [63:0] count,
15 output reg      int_st,
16 output wire     halt_ack,
17 input wire      tdr0_wr_en,
18 input wire      tdr1_wr_en,
19 input wire [31:0] tdr_wr_data,
20 input wire [3:0]  tdr_wr_strb
21 );
22 assign halt_ack = halt_req & dbg_mode;
23
24 reg [7:0] limit;
25 always @(*) begin
26     case (div_val[3:0])
27         4'h1: limit = 8'd1;
28         4'h2: limit = 8'd3;
29         4'h3: limit = 8'd7;
30         4'h4: limit = 8'd15;
31         4'h5: limit = 8'd31;
32         4'h6: limit = 8'd63;
33         4'h7: limit = 8'd127;
34         4'h8: limit = 8'd255;
35         default: limit = 8'd0;
36     endcase
37 end
38
39 reg [7:0] cnt;
40 wire cnt_rst = (cnt == limit) | (!timer_en) | (!div_en);
41 wire [7:0] cnt_pre = halt_ack ? cnt : cnt_rst ? 8'h00 : (cnt + 1'b1);
42
43 always @(posedge sys_clk or negedge sys_rst_n) begin
44     if (!sys_rst_n) cnt <= 8'h00;
45     else cnt <= cnt_pre;
46 end
47
48 wire def_mode      = timer_en & !div_en;
49 wire ctrl_mode_0    = timer_en & div_en & (div_val == 4'h0);
50 wire ctrl_mode_other = timer_en & div_en & (div_val != 4'h0) & (cnt == limit);
51 wire count_en       = (def_mode | ctrl_mode_0 | ctrl_mode_other) & !halt_ack;
52
53 reg [31:0] tdr0, tdr1;
54 assign count = {tdr1, tdr0};
55 wire [63:0] cnt_plus1 = count + 1'b1;
56
57 wire [31:0] tdr0_pre, tdr1_pre;
58 assign tdr0_pre[7:0] = (tdr0_wr_en & tdr_wr_strb[0]) ? tdr_wr_data[7:0] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[7:0] : tdr0[7:0];
59 assign tdr0_pre[15:8] = (tdr0_wr_en & tdr_wr_strb[1]) ? tdr_wr_data[15:8] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[15:8] : tdr0[15:8];
60 assign tdr0_pre[23:16] = (tdr0_wr_en & tdr_wr_strb[2]) ? tdr_wr_data[23:16] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[23:16] : tdr0[23:16];
61 assign tdr0_pre[31:24] = (tdr0_wr_en & tdr_wr_strb[3]) ? tdr_wr_data[31:24] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[31:24] : tdr0[31:24];
62
63 assign tdr1_pre[7:0] = (tdr1_wr_en & tdr_wr_strb[0]) ? tdr_wr_data[7:0] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[39:32] : tdr1[7:0];
64 assign tdr1_pre[15:8] = (tdr1_wr_en & tdr_wr_strb[1]) ? tdr_wr_data[15:8] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[47:40] : tdr1[15:8];
65 assign tdr1_pre[23:16] = (tdr1_wr_en & tdr_wr_strb[2]) ? tdr_wr_data[23:16] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[55:48] : tdr1[23:16];
66 assign tdr1_pre[31:24] = (tdr1_wr_en & tdr_wr_strb[3]) ? tdr_wr_data[31:24] :
timer_en_neg ? 8'h0 : count_en ? cnt_plus1[63:56] : tdr1[31:24];
67
68 always @(posedge sys_clk or negedge sys_rst_n) begin
69     if (!sys_rst_n) begin
70         tdr0 <= 32'h0;
71         tdr1 <= 32'h0;
72     end else begin
73         tdr0 <= tdr0_pre;
74         tdr1 <= tdr1_pre;
75     end
76 end
77
78 always @(posedge sys_clk or negedge sys_rst_n) begin

```

```

79     if (!sys_rst_n) begin
80         int_st <= 1'b0;
81     end else begin
82         if (int_clr_pulse) int_st <= 1'b0;
83         else if (count == tcmp) int_st <= 1'b1;
84     end
85 end
86 assign tim_int = int_st && int_en;
87 endmodule

```

Listing 3: Counter Core Engine (timer_core.v)

4.4 timer_top.v

```

1 module timer_top (
2     input wire      sys_clk,
3     input wire      sys_rst_n,
4     input wire      tim_psel,
5     input wire      tim_penable,
6     input wire      tim_pwrite,
7     input wire [11:0] tim_paddr,
8     input wire [31:0] tim_pwdata,
9     input wire [3:0]  tim_pstrb,
10    output wire [31:0] tim_prdata,
11    output wire      tim_pready,
12    output wire      tim_pslverr,
13    output wire      tim_int,
14    input wire      dbg_mode
15 );
16 wire      error_detected, bus_wr_en, bus_rd_en;
17 wire [3:0] div_val;
18 wire      div_en, timer_en, timer_en_neg;
19 wire [63:0] tcmp, count;
20 wire      int_en, int_clr_pulse, halt_req, int_st, halt_ack;
21 wire      tdr0_wr_en, tdr1_wr_en;
22 wire [31:0] tdr_wr_data;
23 wire [3:0]  tdr_wr_strb;
24
25 apb_slave_interface u_apb_slave (
26     .sys_clk(sys_clk), .sys_rst_n(sys_rst_n),
27     .tim_psel(tim_psel), .tim_penable(tim_penable), .tim_pwrite(tim_pwrite),
28     .tim_pready(tim_pready), .tim_pslverr(tim_pslverr),
29     .error_detected(error_detected), .bus_wr_en(bus_wr_en), .bus_rd_en(bus_rd_en)
30 );
31
32 timer_reg_file u_reg_file (
33     .sys_clk(sys_clk), .sys_rst_n(sys_rst_n),
34     .tim_psel(tim_psel), .tim_penable(tim_penable), .tim_pwrite(tim_pwrite),
35     .bus_wr_en(bus_wr_en), .bus_rd_en(bus_rd_en),
36     .tim_paddr(tim_paddr), .tim_pwdata(tim_pwdata), .tim_pstrb(tim_pstrb),
37     .tim_prdata(tim_prdata), .error_detected(error_detected),
38     .div_val(div_val), .div_en(div_en), .timer_en(timer_en),
39     .timer_en_neg(timer_en_neg), .tcmp(tcmp), .int_en(int_en),
40     .int_clr_pulse(int_clr_pulse), .halt_req(halt_req),
41     .count(count), .int_st(int_st), .halt_ack(halt_ack),
42     .tdr0_wr_en(tdr0_wr_en), .tdr1_wr_en(tdr1_wr_en),
43     .tdr_wr_data(tdr_wr_data), .tdr_wr_strb(tdr_wr_strb)
44 );
45
46 timer_core u_core (
47     .sys_clk(sys_clk), .sys_rst_n(sys_rst_n),
48     .div_val(div_val), .div_en(div_en), .timer_en(timer_en),
49     .timer_en_neg(timer_en_neg), .tcmp(tcmp), .int_en(int_en),
50     .int_clr_pulse(int_clr_pulse), .halt_req(halt_req), .dbg_mode(dbg_mode),
51     .tim_int(tim_int), .count(count), .int_st(int_st), .halt_ack(halt_ack),
52     .tdr0_wr_en(tdr0_wr_en), .tdr1_wr_en(tdr1_wr_en),
53     .tdr_wr_data(tdr_wr_data), .tdr_wr_strb(tdr_wr_strb)
54 );
55 endmodule

```

Listing 4: Top-Level Structural Wrapper (timer_top.v)

5 Verification Plan & Simulation Results

5.1 Verification Plan (V-Plan)

Table 11 lists the verification items defined to test the Timer IP functionality.

Table 11: Comprehensive Verification Plan (V-Plan)

ID	Item	Test Sequence	Pass Condition	Class
1	rst_chk	Assert <code>sys_rst_n = 0</code>	All registers return exact reset default values.	reg_init_chk
2	rd_wr_chk	Write patterns 0x55..., 0xAA..., 0x5A...	Read data matches written value for writable bits.	reg_rw_chk
3	reserved_chk	Write/read unmapped addresses (0x20, 0x100)	RAZ/WI: Reads return 0, writes ignored.	reg_reserved_chk
4	1hot_chk	Write unique signatures across all registers	Readback verifies complete address isolation.	reg_1hot_chk
5	byte_access	Write byte strobes <code>pstrb = 1, 2, 4, 8</code>	Targeted byte lanes update; non-strobed bytes hold.	reg_byte_access
6	cnt_ctrl	Sweep all divider modes $\div 1 \dots \div 256$	Tick frequency matches $2^{\text{div_val}}$. Auto-clear on disable.	cnt_ctrl_chk
7	apb_protocol	Check setup/access phase and wait state	<code>pready</code> asserts with 1-cycle delay. Unselected reads = 0.	apb_protocol_chk
8	multiple_acc	Back-to-back consecutive writes/reads	Continuous transfers complete without hangs or loss.	apb_multiple_access
9	unaligned_acc	Issue unaligned APB address accesses	Handled gracefully with zero data response.	apb_unaligned_chk
10	cnt_counting	Preload near 0xFFFF_FFFF boundary	64-bit rollover from TDR0 into TDR1 verified.	cnt_counting_chk
11	interrupt	Match target, clear TISR, mask TIER	Level interrupt asserts, masks to 0, clears via RW1C.	interrupt_chk
12	cnt_halt	Assert <code>dbg_mode</code> and <code>halt_req</code>	<code>halt_ack = 1</code> , counter freezes, resumes on unhalt.	cnt_halt_chk
13	pslverr	Write <code>div_val > 8</code> or modify while running	<code>tim_pslverr = 1</code> asserts, write data rejected.	apb_pslverr_chk

5.2 Golden Model Regression Results

The synthesizable RTL design was verified using the instructor's Golden Model regression suite (`./run.csh cov`). All ****13/13 testcases passed cleanly with 0 errors****, as shown in Table 12.

Table 12: Golden Model Test Execution Results (TOTAL/PASSED/REMAIN: 13 / 13 / 0)

#	Pattern Name (PAT_NAME)	Result	Sim Time	Verification Scope
1	reg_init_chk	PASSED	475 ns	Reset values across all 8 registers.
2	reg_rw_chk	PASSED	4,075 ns	Multi-pattern R/W data integrity.
3	reg_reserved_chk	PASSED	825 ns	RAZ/WI memory boundaries.
4	reg_lhot_chk	PASSED	1,225 ns	Address line isolation and decoding.
5	reg_byte_access	PASSED	19.3 μ s	Byte-strobe writes (pstrb[3:0]).
6	cnt_ctrl_chk	PASSED	27.4 ms	All prescaler modes and auto-clear.
7	apb_protocol_chk	PASSED	2,075 ns	APB protocol 1-cycle wait states.
8	apb_multiple_access	PASSED	3,250 ns	Consecutive back-to-back transactions.
9	apb_unaligned_chk	PASSED	1,450 ns	Unaligned access handling.
10	cnt_counting_chk	PASSED	69.3 μ s	64-bit counter rollover.
11	interrupt_chk	PASSED	12.8 μ s	Match assertion and RW1C clear.
12	cnt_halt_chk	PASSED	27.1 ms	Debug halt freeze and timing preservation.
13	apb_pslverr_chk	PASSED	3,100 ns	Error handling on prohibited accesses.

```

apb_pslverr_chk
-----
| PAT_NAME | RUN DATE | RESULT |
|-----|-----|-----|
| reg_init_chk | 22:18:12 Aug 27 2026 | PASSED |
|-----|-----|-----|
| reg_rw_chk | 22:18:15 Aug 27 2026 | PASSED |
|-----|-----|-----|
| reg_reserved_chk | 22:18:18 Aug 27 2026 | PASSED |
|-----|-----|-----|
| reg_lhot_chk | 22:18:20 Aug 27 2026 | PASSED |
|-----|-----|-----|
| reg_byte_access | 22:18:23 Aug 27 2026 | PASSED |
|-----|-----|-----|
| cnt_ctrl_chk | 22:18:43 Aug 27 2026 | PASSED |
|-----|-----|-----|
| apb_protocol_chk | 22:18:45 Aug 27 2026 | PASSED |
|-----|-----|-----|
| apb_multiple_access | 22:18:47 Aug 27 2026 | PASSED |
|-----|-----|-----|
| apb_unaligned_chk | 22:18:49 Aug 27 2026 | PASSED |
|-----|-----|-----|
| cnt_counting_chk | 22:18:51 Aug 27 2026 | PASSED |
|-----|-----|-----|
| interrupt_chk | 22:18:53 Aug 27 2026 | PASSED |
|-----|-----|-----|
| cnt_halt_chk | 22:19:12 Aug 27 2026 | PASSED |
|-----|-----|-----|
| apb_pslverr_chk | 22:19:14 Aug 27 2026 | PASSED |
|-----|-----|-----|
TOTAL /PASSED/REMAIN: 13/13/0

```

Figure 22: Test Cases Simulation Execution Report

5.3 Code Coverage Report

Code coverage was collected across all design units using QuestaSim The synthesizable hardware design achieves **91.59

Table 13: Design Unit Code Coverage Metrics (from test.ucdb)

Design Unit / Module	Statements	Branches	Toggles	Total Coverage
apb_slave_interface	100.00% (8/8)	100.00% (2/2)	100.00% (4/4)	95.00%
timer_reg_file	100.00% (45/45)	96.00% (48/50)	98.16% (214/218)	98.06%
timer_core	85.00% (34/40)	90.00% (63/70)	90.09% (400/444)	83.02%
Top: u_dut (timer_top)	93.54% (87/93)	92.62% (113/122)	94.75% (976/1030)	91.59%

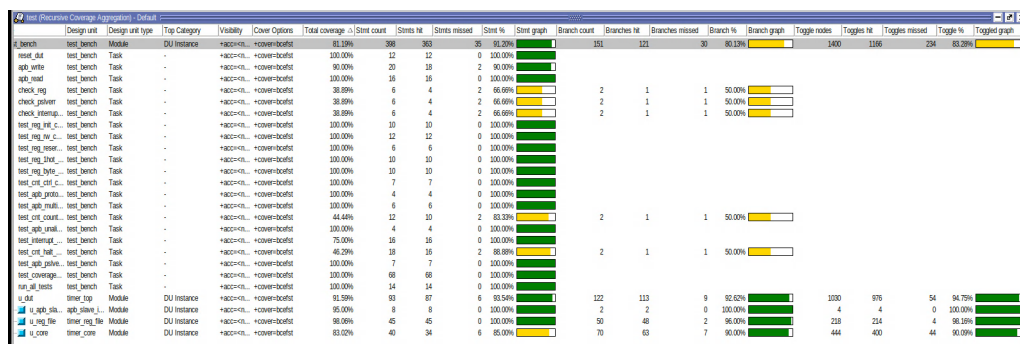


Figure 23: Code Coverage Report Summary

6 Conclusion

The Advanced 64-bit Timer IP (**timer_top**) has been successfully implemented and verified. The synthesizable RTL satisfies all functional requirements, including multi-mode prescaling, APB slave wait states, byte-strobe access, error detection, debug-halt preservation, and interrupt handling.

Verification closure was achieved through:

- 100% Pass rate on the internal 69-item self-checking testbench regression suite.
- 100% Pass rate on the instructor's Golden Model regression suite (13 / 13 / 0).
- 91.59% Total code coverage in hardware design units.